Gizmonic Cambot: mechanical specification

PETG housing, retainer, modular arms and dual-magnet foot · W-Mount contract

Gizmonic Cambot project · open hardware (CERN-OHL-P-2.0)
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1. Purpose and deliverables

Document the 3D-printable enclosure and modular mount for the Cambot camera on the FORIOT ESP32-S3-CAM board, and keep the **W-Mount** interface stable so community feet and arms can mate with it.

Source of truth for printable files: the frozen release under `hardware/cambot-minimum-petg-kit/` (release **2026-11-04**). That kit ships STL, STEP and 3MF exports checked by `manifest.json`, CadQuery generators under `source/`, and a kit README with PETG settings, hardware and assembly. Edit the generators, regenerate, and update exports and checksums together.

Format	Purpose
CadQuery Python under `source/`	Parametric edits
STEP	CAD interchange
STL and oriented 3MF	Printing
Plate 3MFs in `plates/`	Grouped print layouts
Kit `README.md` + `LICENSE.md`	Print, assembly, provenance

Units are millimeters. Tolerances not stated are ±0.2 mm on the printed part.

> **Dimensions marked TBV (to be verified)** come from vendor listings or clone boards, not from calipers. Before freezing geometry that depends on them, measure a physical board. Every TBV value should stay a top-level parameter in the generators.

2. System overview (current kit)

The minimum PETG kit is an **open housing** with modular W-Mount arms and a dual external-magnet foot. It deliberately omits a front cover, screw clamp and embedded-magnet puck.

Part	Role
`housing_v3`	Drop-in body, two center posts, header clearance, camera-side `knuckle2`
`retainer_v3_short`	Short bridge retainer that latches over the PCB
`rod_male_25` / `rod_male_50`	Two-finger arm with 12 mm spigot
`rod_female_25_6-32` / `rod_female_50_6-32`	Three-finger arm with bore and 6-32 locking pilot
`dual_magnet_foot`	Three-finger base; magnets mount under the plate against steel

Only five pieces assemble into one mount: housing, retainer, one male rod, one female rod, and the foot. The unused rod lengths are alternatives.

Male + female shafts	Approx. pivot-to-pivot
25 + 25	92–96 mm
25 + 50 (or 50 + 25)	117–121 mm
50 + 50	142–146 mm

These are pivot distances, not steel-to-lens reach. Reach also depends on joint angles and magnet thickness.

The housing and feet meet at the **W-Mount** (section 6). That interface is the stable contract. Publish `knuckle2` / `knuckle3` as standalone modules (the kit already does this in `w_mount.py`).

3. Reference board: FORIOT ESP32-S3-CAM

Orientation used throughout: hold the board component side toward you, with the **antenna end at the top** and the **two USB-C ports on the bottom edge**. “Front” is the component and camera side; “rear” is the side with the microSD slot and header pins.

Feature	Value	Status
PCB outline	62.6 × 28.3 mm	Vendor figure, TBV
PCB thickness	1.6 mm	TBV
Mounting holes	None. Kit retains on two center posts + retainer.	Confirmed in kit
Header rows	2 × 20 pins, 2.54 mm pitch, rows 25.4 mm apart	Vendor figure
Header pins on rear (if soldered)	Up to 8.5 mm below the PCB rear face	Housing clears headers
Tallest front component, excluding the camera	ESP32-S3-WROOM-1 shield, ~3.2 mm	TBV
Antenna keep-out	Top ~6 mm of the PCB (module antenna)	Retainer opening ~24 mm for ~17.7 mm antenna width; fit still TBV
USB-C ports	Two, on the bottom edge. Label TTL (CH340) and OTG.	Centers TBV
USB-C plug overmold clearance	12.5 × 7.0 mm per plug, minimum	Typical cable
Buttons	RST and BOOT near the USB edge, front side	Open housing; accessible
LEDs	Power, TX, WS2812 RGB (GPIO 48), user LED (GPIO 2)	Open housing
microSD	Push-push slot on rear; card insertion edge	Access with housing open / unmounted
Camera	OV3660 module on a 24-pin 0.5 mm flex, lens on the front	Module size, lens Ø and flex length TBV

Camera module (TBV): typical modules are about 8.5 × 8.5 mm with an M8-class lens barrel about 8 mm in diameter. Wide-angle variants (120° and 160°) of the OV3660 fit the same connector. A future front cover (section 8) must work with lens fields of view from 66° to 160°.

4. Design rules for all parts

Rule	Value
Target printers	0.4 mm nozzle, 0.2 mm layers, bed ≥ 180 × 180 mm
Materials	PETG (default). PLA allowed for indoor prototypes.
Minimum wall	1.6 mm (four perimeters at 0.4 mm)
Sliding fit	Arms: 12 mm spigot into 12.4 mm bore, ≥ 20 mm engagement

Rule	Value
Overhangs	Support as noted in the kit README; finger faces parallel to the bed
Pivots	M5 bolt through W-Mount fingers with an M5 nut
Arm lock	6-32 machine screw into female-rod pilot; bears on male spigot
Magnets	External rubber-coated assemblies under the foot (M4), not glued discs
Metal near antenna	No inserts, nuts or magnets within 15 mm of the antenna end

Starting PETG profile from the kit: 5 walls, 6 top/bottom layers, 40–50% infill. Finer layers alone will not correct an interfering fit.

5. Body (current kit)

5.1 Housing (`housing_v3`)

- Open drop-in body. The PCB sits on **two center posts**; there is no edge sled and no front cover in this release.
- Header clearance on the rear side of the cavity.
- Camera-side **two-finger W-Mount** (`knuckle2`) for the arm stack.
- Print orientation: on its side in the plate 3MF; support tray and mounting overhangs; clean the latch windows.

5.2 Retainer (`retainer_v3_short`)

- 4.5 mm thick bridge with a shortened span (7.3 mm removed relative to the earlier long retainer). Bridge inner edge is about 0.5 mm past the case end.
- Latches outward (not inward) over the housing.
- 24 mm antenna opening. Longitudinal clearance still needs a test fit on each board.
- Print flat; preserve latch clearances; avoid support in those gaps.

5.3 Assembly of board into housing

1. Clean and test-fit the empty housing and retainer.
2. Drop the PCB onto the two center posts (antenna end toward the retainer opening).
3. Seat the retainer. Check antenna, header, camera ribbon and USB clearance.
4. Insert the microSD card before or after seating, as access allows.

No M2 case screws are used in the minimum kit.

6. W-Mount interface (fixed contract)

The W-Mount follows the de facto GoPro finger mount, so off-the-shelf GoPro accessories work. **Check the final parameters against a genuine GoPro buckle and adjust until both mate with a firm, wobble-free fit.**

Parameter	Default	Notes
`finger_t`	3.0	Finger thickness
`finger_gap`	3.2	Gap between fingers
`finger_r`	7.5	Outer radius around the pivot
`finger_reach`	9.0	Pivot center to base surface
`bolt_d`	5.3	Clearance for M5
Camera / male-rod side (`knuckle2`)	2 fingers	Total width 9.2
Foot / female-rod side (`knuckle3`)	3 fingers	Total width 15.4. One outer finger has a captive M5 hex-nut pocket.
Bolt	M5 × 25 thumbscrew or socket-head bolt	Verify reach with your hardware

Print fingers so the layers stack across the finger thickness (finger faces parallel to the bed). This avoids layer splits at the pivot. Fillet the finger roots with r = 1.5.

The kit modules live in each generator's `w_mount.py` . Keep them consistent across housing, arms and foot.

7. Arms and foot

7.1 Modular arms

Feature	Spec
Shaft lengths	25 mm and 50 mm (male and female)
Spigot / bore	Ø12 mm spigot into Ø12.4 mm bore
Engagement	Minimum 20 mm; groove must be fully hidden (20–24 mm insertion)
Lock	6-32 × ~3/8 in machine screw in female pilot; forms thread in PETG by hand
Joints	Male rod: `knuckle2` . Female rod: `knuckle3` .

There is no positive pullout catch; the set screw bears on the spigot. Intended for occasional repositioning, not continuous adjustment under load.

Print orientation. Horizontal with finger faces on the bed. Support round undersides and projecting fingers; clear the bore before assembly.

7.2 Dual-magnet foot (`dual_magnet_foot`)

Feature	Spec
Plate	~32 mm wide, 6 mm thick, reinforced joint roots
Magnet spacing	56 mm centers
Magnets	Two rubber-coated mounting magnets, nominal Ø22 mm with M4 stud/thread (25–30 mm also fit; 32 mm max without overhang)
Mounting	Magnets under the flat base, magnetic face against steel; M4 washers (~12 mm OD) in top recesses with metal fasteners
Peel lip	Lever the foot off steel at the lip; do not pull on the rods
Knuckle	`knuckle3` on the top

Reference magnet class: rubber-coated outdoor magnetic assembly base, ~8 lb pull rating (manufacturer figure; not a validated arm working load). Confirm stud length / thread depth before choosing nuts vs screws; do not crush the 4 mm of PETG under each washer recess.

Print orientation. Long side edge, finger faces parallel to the bed; brim and support on projecting fingers. STEP is assembly orientation; use 3MF/STL for print orientation.

7.3 Hardware for one assembled unit

- 2 × M5 × 25 bolts or thumbscrews and 2 × M5 nuts (end pivots)
- 1 × 6-32 machine screw, ~3/8 in (arm lock)
- 2 × rubber-coated magnets with M4 mounting
- 2 × M4 washers (~12 mm OD, ~1 mm thick) plus matching M4 nuts or screws

8. Future / deferred parts

The October design brief called for a board kit (`sled` , `front_cover`), a universal rear shell with SD door, a screw clamp foot and a glued-in magnet puck. Those are **not** in the 2026-11-04 kit. Keep them as optional follow-ons:

Deferred part	Intent
Front cover + lens window	Optics protection, LED baffles, button pins
Edge sled	Alternate retention for boards without post fit

Deferred part	Intent
Screw clamp foot	Shelf / extrusion clamp with printed or steel screw
Embedded-magnet puck	Slide-in Ø10 discs glued behind a thin skin
Foot template	Minimal `knuckle3` plate for community feet

If revived, they must mate with the same W-Mount contract in section 6 and must not break the current housing knuckle geometry.

9. Assembly (minimum kit)

1. Test-fit housing and retainer; seat the PCB and retainer (section 5.3).
2. Fasten the magnets under the foot with washers and M4 hardware.
3. Join foot `knuckle3` to male-rod `knuckle2` with an M5 bolt and nut.
4. Form the 6-32 thread by hand in the empty female bore, back the screw out, insert the male spigot to full engagement, set roll, tighten lightly.
5. Join female-rod `knuckle3` to housing `knuckle2` with the second M5 bolt.
6. On vertical steel, orient the magnets to resist the arm's tipping moment. Remove by peeling at the foot lip.

Test the complete unit at full extension over a padded surface, including cable pull. Check slipping, sag, latch seating and finger roots. Do not force binding joints.

10. Acceptance checks

#	Check	Pass criterion
1	Board fit	PCB seats on posts; retainer latches; USB-C plugs seat; antenna clears opening
2	Ribbon / headers	Camera flex and header pins clear the housing
3	W-Mount	Mates with printed arms/foot and a genuine GoPro buckle; holds tilt hand-tight
4	Arm lock	Spigot reaches min engagement; 6-32 holds roll without stripping on first sets
5	Magnet	Holds on vertical steel at full extension; peel lip works; PETG under washers uncrushed

#	Check	Pass criterion
6	Printability	Every kit part prints from the plate 3MF orientations in PETG at 0.2 mm
7	Wi-Fi	RSSI at 5 m is no worse than 6 dB below the bare board with the foot on steel
8	Checksums	`python source/verify_release.py` passes for the frozen exports

11. Open questions

1. Confirm remaining TBV board dimensions from a physical board, especially lens position, USB-C centers and antenna envelope vs the retainer opening.
2. Validate long-term PETG creep on the arm lock and magnet washer stack.
3. Decide whether a front cover is worth adding for dust and lens protection without blocking Wi-Fi or overheating.
4. If a clamp foot returns, prefer M5 W-Mount compatibility over a one-off interface.